EX NO:11 **Node.js and Express.js with Array**

DATE:

Problem Understanding:

The problem is to develop a Student Management REST API using Node.js and Express.js where student records are stored in an array. The application should allow users to add, retrieve, update, and delete student details using appropriate HTTP methods and REST API endpoints, with the APIs tested through Postman.

Problem Statement:

Develop a REST API using Node.js and Express.js to manage student records stored in an array. The application should support adding, retrieving, updating, and deleting student records using appropriate HTTP methods and REST API endpoints.

Student Management REST API

App.jsx

```
import { useState } from "react";

function App() {
  const [students, setStudents] = useState([
    { id: 101, name: "Chotta", department: "CSE", year: 2 },
    { id: 102, name: "Bheem", department: "ECE", year: 3 },
    { id: 103, name: "Raju", department: "CSE", year: 2 },
    { id: 104, name: "Jaggu", department: "AIDS", year: 3 }
  ]);
  const [studentName, setStudentName] = useState("");
  const [studentDept, setStudentDept] = useState("");
  const [studentYear, setStudentYear] = useState("");
  const addStudent = () => {
    if (
      studentName === "" || studentDept === "" || studentYear === ""
    ) {
```

```
    return;
  }
  const newStudent = {
    id: students.length + 101,
    name: studentName,
    department: studentDept,
    year: studentYear
  };
  setStudents([...students, newStudent]);
  setStudentName("");
  setStudentDept(""); setStudentYear("");
};
const removeStudent = (studentId) =>
  { const updatedList =
  students.filter( student => student.id !==
  studentId
  );
  setStudents(updatedList);
};
return (
  <div>
    <h1>College Student Records</h1>
    <input
      type="text"
      placeholder="Enter Name"
      value={studentName}
      onChange={(e) => setStudentName(e.target.value)}
    />
```

```
<input
  type="text"
  placeholder="Enter Department"
  value={studentDept}
  onChange={(e) => setStudentDept(e.target.value)}
/>
<input
  type="number"
  placeholder="Enter Year"
  value={studentYear}
  onChange={(e) => setStudentYear(e.target.value)}
/>
<button onClick={addStudent}>
  Add Record
</button>
<h2>Student Records</h2>
{students.map((student) => (
  <div key={student.id}>
    <p>
      <b>ID:</b> {student.id}
      <br />
      <b>Name:</b> {student.name}
      <br />
      <b>Department:</b> {student.department}
      <br />
      <b>Year:</b> {student.year}
    </p>
    <button onClick={() => removeStudent(student.id)}>
```

```
                Remove
              </button>
              <hr />
            </div
          ))}
        </div>
      );
    }
    export default App;
```

Server.js

```
const express = require("express");
const app = express();
app.use(express.json());
let students = [
    { id: 101, name: "Sarathi", department: "CSE", year: 2 },
    { id: 102, name: "Kamal", department: "ECE", year: 3 },
    { id: 103, name: "Bala", department: "CSE", year: 2 },
    { id: 104, name: "Sundar", department: "AIDS", year: 3 }
];
app.get("/api/students", (req, res) =>
    { res.json(students);
});
app.get("/api/students/:id", (req, res) =>
    { const id = Number(req.params.id);
    const student = students.find(s => s.id === id);
    if (!student) {
        return
            res.status(404).json({ message:
            "Student does not exist"
```

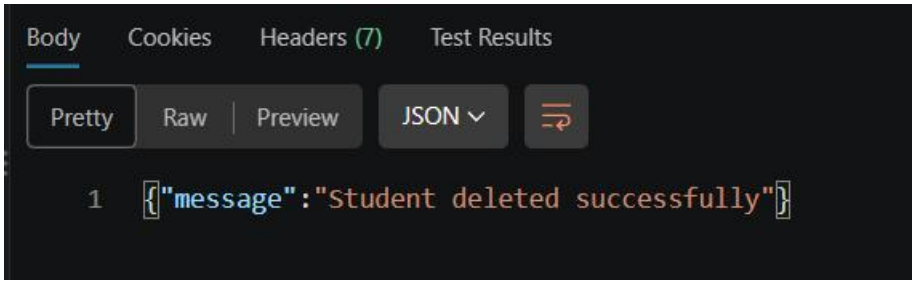
```
    });}
  res.json(student);
});
app.post("/api/students", (req, res) =>
  { const newStudent = {
    id: students.length + 101,
    name: req.body.name,
    department: req.body.department,
    year: req.body.year
  };
  students.push(newStudent);
  res.status(201).json({
    message: "Student added successfully",
    student: newStudent
  });
});
app.put("/api/students/:id", (req, res) =>
  { const id = Number(req.params.id);
  const student = students.find(s => s.id === id);
  if (!student) {
    return
      res.status(404).json({ message:
      "Student does not exist"
    });
  }
  student.name = req.body.name;
  student.department = req.body.department;
  student.year = req.body.year;
  res.json({
```

```
        message: "Student updated successfully",
        student: student
    });
});
app.delete("/api/students/:id", (req, res) =>
    { const id = Number(req.params.id);
    const oldLength = students.length;
    students = students.filter(s => s.id !== id);
    if (students.length === oldLength) {
        return
            res.status(404).json({ message:
            "Student does not exist"
        });
    }
    res.json({
        message: "Student deleted successfully"
    });
});
app.listen(3000, () => {
    console.log("Server started at http://localhost:3000");});
```

Output:







Problem Understand & Approach (10)	Implementation and Execution (15)	Tool/ Technology usage (10)	Code Quality & Documentation (5)	Viva (5)	Time Management (5)	Total (50)

RESULT:

Thus, the Student Management REST API was successfully developed using Node.js and Express.js with an array. The application implemented GET, POST, PUT, and DELETE operations for adding, retrieving, updating, and deleting student records through RESTful API endpoints.